CCCA Cell-State Analysis — Zhang et al. 2019, Hematologic

Follicular Lymphoma / transformed DLBCL · 10x · 5,645 malignant B cells (4 samples, 2 patients)

GO / NO-GO GATES

Gate 1 — mixing & continuity GO

Harmony(theta=10) integrates all 4 samples + both patients (FL1018/FL2001): per-neighbourhood iLISI 1.05 -> 1.88 of a 3.11 perfect-mixing ceiling. One connected component (0 islands removed). Judged on the regenerated embedding + per-neighbourhood iLISI across a theta sweep (theta=1 archipelago -> theta=10 mixed), not a scalar: mixed AND continuous.

Gate 2 — module geography GO

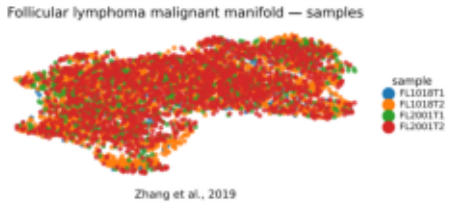
Non-branching case: a single Gast -> Crest developmental gradient. Absolute module expression is weak (raw means ~0.01, control-pool level). The Gast->Crest axis is spatially organized and confound-independent: along pseudotime Gast falls (rho -0.51, survives regressing out progression+interferon+proliferation, R2=0.07 -> -0.46) and Crest rises to a terminal gradient (rho +0.32, R2=0.16 -> +0.38). pSom is the exception — ~63% of its variance is confound-explained and its high-decile is 54% cycling (vs 14% baseline), so pSom is substantially proliferation-confounded, not an independent axis (not required by the non-branching criteria). Gast<->pSom mild cross-repression (r=-0.18, no co-expression hub); Crest opposes both Gast (r=-0.54) and pSom (r=-0.60) -> inverse terminal gradient, no Gast-arm inversion (flag = False).

Gate 3 — cross-repression NA

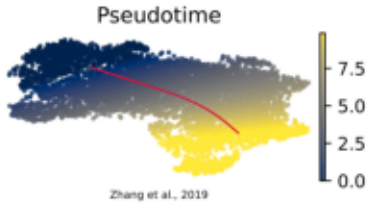
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks robust across PPT Nodes=15/20/30 and every curve fit; no Gast/pSom co-expression hub). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks are not produced, per the deliverable spec).

STEP FIGURES

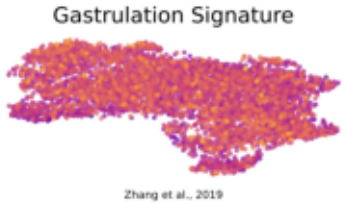
01 · Samples (Gate 1)



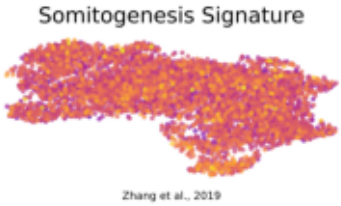
02 · Pseudotime



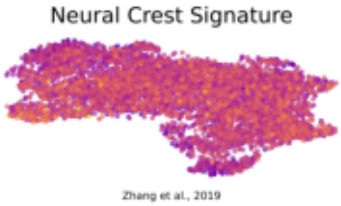
03 · Gastrulation



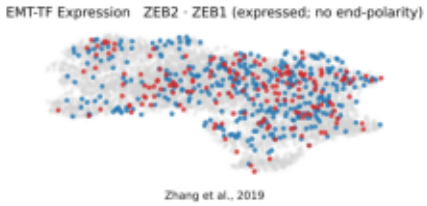
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs



Dataset: Follicular lymphoma / transformed DLBCL (Zhang 2019), 10x, 5,645 malignant B cells over 4 samples (2 patients, each primary + transformed/progressed); Gast-high developmental origin -> Crest-high terminal gradient on one continuous manifold.

Verdict: Pattern reproduces in the non-branching form (Linear): one continuous gastrulation -> neural-crest axis, Crest terminal, mild Gast/pSom cross-repression. G1 GO / G2 GO / G3 NA. Gast & Crest confound-independent; pSom proliferation-confounded. EMT-TFs ZEB2/ZEB1 expressed but sparse and not end-polarized.

PARAMETER CHOICES

Sample threshold	>=100 cells (floor; no dominant cliff, all 4 samples 680-2405) -> 4/4 kept, 5,645 cells
Malignant selection	publication cell_type == 'Malignant' (5,645 B cells); normal T/B compartment dropped
Overdispersed genes	636 (scFates find_overdispersed); no sc.pp.scale before PCA (per protocol)
Integration	Harmony on sample, theta=10 (iLISI peaks at 10, declines at 20/40; strongest without collapse)
Embedding	n_pcs=30, n_neighbors=30, UMAP spread=1.3 (mid robust 20-40 range); 0 islands removed
Geometry	Non-branching -> scFates principal curve (ElPiGraph), Nodes=20, epg_lambda=0.3, epg_mu=100
Topology check	2 tips / 0 forks (robust); fold self-approach 2.97 (>1 fold-free); continuity max|corr(t,DC)|=0.74
Root	curve tip = Gast/pSom-high, Crest-low developmental origin (derived from signature order)
Module means (raw)	Gast +0.012 · pSom +0.006 · Crest +0.009 (weak, control-pool level; read off geography)
EMT-TFs shown	ZEB2 (27%) · ZEB1 (12%); detection-gated (>3%); SNAI/TWIST/FOXC/PRRX near-absent